

HW 8: RMS with Resources

Schedulability tests

a)

- $T_1 = (5; 15; 2; 2)$
- $T_2 = (2; 10; 1; 1)$
- $T_3 = (7; 23; 1; 2)$

$$T_1 = \frac{5}{15} = \frac{1}{3}$$

$$T_2 = \frac{2}{10} = \frac{1}{5}$$

$$T_3 = \frac{7}{23}$$

$$T_* = \frac{1}{3} + \frac{1}{5} + \frac{7}{23} \approx 0.8377$$

$$U \leq 3(2^{1/3} - 1)$$

$$U \leq 0.7798$$

$$0.8377 \not\leq 0.7798$$

$\Rightarrow$ Not confirmed to be Schedulable

b)

- $T_1 = (4; 14; 1; 2)$
- $T_2 = (2; 10; 0; 0)$
- $T_3 = (7; 30; 3; 4)$

$$T_1 = \frac{4}{14} = \frac{2}{7}$$

$$T_2 = \frac{2}{10} = \frac{1}{5}$$

$$T_3 = \frac{7}{30}$$

$$T_* = \frac{2}{7} + \frac{1}{5} + \frac{7}{30} \approx 0.7190$$

$$U \leq 3(2^{1/3} - 1)$$

$$0.7190 \leq 0.7798$$

⇒ Definitely Schedulable

Schedules

Bitte denken Sie sich überall die "!" hinter die Deadlines dazu, die habe ich vergessen. Die Deadlines konnntent überall getroffen werden, wenn auch teilweise sehr knapp (A tic 22).

Table A

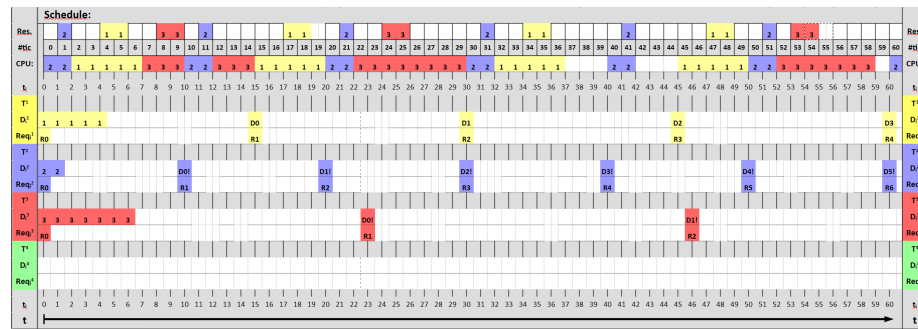


Table B

